

## SHENG Dong 盛东

Doctoral candidate at School of Engineering, Westlake University, China.

Visiting student at Agroscope, Switzerland.

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### RESEARCH INTEREST

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Plastic Pollution, Climate Change, Pollination, Ecosystem Services, Artificial Intelligence, Computer Vision, Complex Systems, Modelling, Science-Policy Interface

### EDUCATION

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- PhD candidate** in Environmental Sciences & Engineering 2023 Sept – present  
School of Engineering, Westlake University (joint program with Zhejiang University)  
Supervisors: Thomas Cherico WANGER & LI Ling (李凌)
- Visiting PhD student** 2025 Oct – present  
Department of Plant Production Systems, Agroscope  
Supervisor: Thomas Cherico Wanger
- Bachelor of Science** in Environmental Sciences 2019 Sept – 2023 Jun  
College of Environmental & Resource Sciences, Zhejiang University  
General GPA = 3.81/4.00  
Supervisor: HUANG Yili
- Minor** in Advanced Honor Class of Engineering Education 2020 Sept – 2023 Jun  
Chu Kochen Honors College, Zhejiang University

### MEMBERSHIPS

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- Student Member – The Ecological Society of America 2024 – present
- Core Member – The Scientists' Coalition for an Effective Plastics Treaty 2025 – present

### PUBLICATIONS

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- Xu, W., Barzegar, S. G., **Sheng, D.**, Toledo-Hernández, M., Lan, Z., & Wanger, T. C. (2025). Identifying Cocoa Flower Visitors: A Deep Learning Dataset. *Scientific Data*, 12(1), 1309. <https://doi.org/10.1038/s41597-025-05631-3>
- Sheng, D.**, Jing, S., He, X., Klein, A.-M., Köhler, H.-R., & Wanger, T. C. (2024). Plastic pollution in agricultural landscapes: An overlooked threat to pollination, biocontrol and food security. *Nature Communications*, 15(1), 8413. <https://doi.org/10.1038/s41467-024-52734-3>
- Darras, K. F. A., Balle, M., Xu, W., Yan, Y., Zakka, V. G., Toledo - Hernández, M., **Sheng, D.**, Lin, W., Zhang, B., Lan, Z., Fupeng, L., & Wanger, T. C. (2024). Eyes on nature: Embedded vision cameras for terrestrial biodiversity monitoring. *Methods in Ecology and Evolution*, 2041-210X.14436. <https://doi.org/10.1111/2041-210X.14436>
- Sheng, D.**, Chen, M., Chen, Q., Huang, Y. (2022). Opposite selection effects of nZVI and PAHs on bacterial community composition revealed by universal and sphingomonads-specific 16SrRNA primers. *Environmental Pollution*, 311, 119893. <https://doi.org/10.1016/j.envpol.2022.119893>

**CONFERENCES & PRESENTATIONS**

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- The 20<sup>th</sup> National Environmental Conference for Doctoral Students, Tsinghua University, Beijing, China. (2026, May 25). Title: *Microplastics in farmlands: A model of conditional fragmentation and source estimation*.
- The 2024 Annual Meeting of the Ecological Society of America, Long Beach, California, United States. (2024, August 8). Title: *Long-term exposure to microplastics and heat affects bumblebee colony development, behavior patterns and social networks*.
- Westlake University Interdisciplinary Frontier Forum on Sustainable Development and Environmental Sciences, Hangzhou, Zhejiang, China. (2024, March 31). Title: *Nano/microplastic effects in agricultural landscapes: an overlooked threat to pollination, biological pest control and food security*.

**PATENTS**

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- Wanger, T. C., **Sheng, D.**, Wang, L., Yuan, X. (2023). *Detection method and apparatus, object monitoring system, computing device, and storage medium* (CN 120124651 A; US 20250190729 A1). China National Intellectual Property Administration, 2025-06-10; United States Patent and Trademark Office, 2025-06-12.

**SELECTED HONORS & AWARDS**

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| China National Scholarship – Ministry of Education of the P.R.C.                                                      | 2025 |
| Dean's Medal – School of Engineering, Westlake University                                                             | 2025 |
| Outstanding Graduate – Westlake University                                                                            | 2025 |
| Award of Honor for Graduate – Zhejiang University                                                                     | 2024 |
| Exploration Award   Westlake University                                                                               | 2024 |
| Certificate of Chu Kochen Honors Program – Zhejiang University                                                        | 2023 |
| The 18th Zhejiang Province "Challenge Cup" Academic and Technological Competition for College Students (Silver Award) | 2023 |
| Interdisciplinary Contest in Modeling (Honorable Mention) – The Consortium for Mathematics and Its Application        | 2022 |
| Zhejiang University – Jiande Scholarship (Second Prize)                                                               | 2021 |
| Zhejiang Province Physics Innovation (Theoretical) Competition for College Students (1 <sup>st</sup> Prize)           | 2020 |
| Zhejiang University Scholarship (Second Prize)                                                                        | 2019 |

**OTHER EXPERIENCES**

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| Visiting student   SASE Lab, Westlake University                             | 2022 May – 2023 Aug  |
| Bioinformatics analysis (supervised by Dr. Yili Huang)   Zhejiang University | 2021 Nov – 2022 Sept |
| Internship   Zhejiang Ecology & Environment Design and Research Institute    | 2022 Jul – 2022 Sept |
| Summer session   University of California, Berkeley                          | 2021 Aug             |

**SKILLS**

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- Software: Python, MATLAB, R, C/C++, CAD, SolidWorks, SketchUp
- Experiment: Raman spectrum, optical microscope, basic chemical/biological experiments
- Data analysis: machine learning, computer vision, meta-analysis